Ritesh Kumar

Jamshedpur, Jharkhand
+91-6206269895 | riteshkumar90359@gmail.com | linkedin.com/in/riteshkumar-tech
github.com/RiteshKumar2e | riteshkr.info | ORCID

Education

Arka Jain University, Jamshedpur **Aug 2022 – May 2026**
B.Tech in Computer Science & Engineering — CGPA: 8.47 *Jamshedpur, Jharkhand*

Experience

TechMantra Global — Web Development Intern **May 2024 – Jul 2024**
Remote — Noida
- Developed the backend for a **recipe discovery app** with **5 REST endpoints** for search, lookup, and auth *(Node.js, Express)*.
- Architected a **hybrid search layer** that queries an **in-memory catalog**, merges it with the external TheMealDB API, and **caps results at 50 per query**; lookups check local data first and **fall back** to the external API.
- Secured the API with **stateless JWT auth** (1-hour expiry), **bcrypt** hashing, duplicate-account checks, and email/phone login.
- Built the **React** frontend with **React Router** (7 pages), **protected routes** for dashboard and profile, and a **Context API** auth store that keeps users logged in across reloads *(React 19, Vite)*.

Machine Vision & Intelligence Lab — Research Intern **May 2025 – Present**
NIT Jamshedpur | Steel defect classification on NEU-DET & SteelDefectX (1,800 + 1,631 images) | GitHub
- **AMFF-CNN (MobileNetV2 + FPN):** Designed a 3.4M-parameter CNN that fuses multi-scale FPN features using channel/spatial attention and learned cross-scale weights; **92.95% macro-F1** on a leakage-free held-out split, Grad-CAM weak localization, 118 ms CPU inference; 6-variant ablation across 3 seeds *(TensorFlow)*.
- **TinySteelLLM (Transformer, from scratch):** Converted ∼30 handcrafted OpenCV descriptors (edges, GLCM texture, FFT, region stats) into short text prompts and classified them with a **2-layer, 4-head Transformer encoder** trained from random initialization — no pretrained weights or external APIs *(PyTorch)*.
- **SteelSense-BiLSTM:** Scaled this to 92 descriptors tokenized into ∼94 tokens and a **2-layer BiLSTM** with attention/max/mean pooling (1.75M parameters); **99.67% ± 0.23 on NEU-DET** and **99.02% ± 0.40 on SteelDefectX** (5 seeds), benchmarked against XGBoost, Random Forest, SVM, MobileNetV3, ShuffleNetV2; first-author paper under review.

Projects

QuickFix: AI-Powered Complaint Management System | Live | GitHub **Dec 2025**
- Launched a support platform where an **async orchestrator** runs category, sentiment, and priority agents **in parallel** on each complaint, then drafts a policy-grounded reply using **RAG**; handles English, Hindi, and Hinglish *(FastAPI, React 19, SQLAlchemy, Turso)*.
- Devised a **fault-tolerant 3-tier LLM fallback** — Groq LLaMA 3.3 → Gemini 2.0 Flash (multi-key rotation) → local TF-IDF/TextBlob — so complaints still get processed when a cloud API fails or hits its rate limit.
- Introduced an **in-memory cache** for repeated queries, cutting API response time from **2.5 s to ∼0.5 s**; guarded the **REST APIs** with Google OAuth 2.0 + email OTP and **JWT-based RBAC**.

Community AI: Multilingual Government Scheme Assistant | Live | GitHub **Jan 2026**
- Delivered a platform used by **100+ active users** to find **50+ government schemes**, courses, and jobs, with an AI assistant (Amazon Q → Groq → Gemini fallback) answering in **5 Indian languages** *(FastAPI, React, SQLAlchemy, Turso)*.
- Created an **automated content scanner** that pulls new schemes and courses from **2 external APIs** (News, YouTube), adds them to the platform, and emails summaries via Brevo.
- Implemented **27 REST endpoints** with **async I/O**, **background tasks** for email, **connection pooling**, and indexed queries, plus JWT + bcrypt auth, Pydantic validation, and **rate limiting**.

SOEIT Student Achievement Portal | Live | GitHub **Feb 2026**
- Shipped the achievement portal for Arka Jain University's School of Engineering & IT: **students upload certificates, faculty verify them, admins manage users** and view analytics *(Node.js, Express 5, React 19, Turso)*.
- Automated **category-based point scoring** that feeds a **weekly leaderboard**, a hub of **90+ curated hackathons**, and **one-click ATS resume export** (DOCX/PDF).
- Optimized the **stateless REST backend** with **SQL indexes** for sub-100 ms lookups on 10k+ records, plus JWT-based RBAC, BLOB storage for certificates, and a React Native companion app.

Technical Skills

Programming: Python, JavaScript, SQL
Software Engineering: Data Structures & Algorithms, OOP, REST APIs, API Integration
Backend: FastAPI, Node.js, Express.js, SQLAlchemy, JWT, OAuth 2.0, RBAC
Databases: PostgreSQL, MySQL, SQLite (Turso), MongoDB, Schema Design, Indexing, Connection Pooling
Reliability & Automation: Async I/O, Caching, Fault Tolerance, Rate Limiting
Tools & Deployment: Git, GitHub, Docker, Render, Vercel
AI/ML: PyTorch, TensorFlow, OpenCV, scikit-learn, RAG, LLM APIs, NLP

Publications & Achievements

Journal (under review): SteelSense-BiLSTM, **first author** with K. K. Singh — *Discover Computing (Springer Nature), 2026*.
Journal (under review): "Lightweight Steel Surface Defect Classification with Multi-Scale Feature Fusion and Post-hoc Weak Localization" — *Journal of Electronic Imaging (SPIE), 2026*.
Conference (presented, co-author): "A Local Feature-Prompt Transformer Framework for Steel Surface Defect Classification & Localization using LLM-Based Steel Surface Defect Detection" — *MAI-2026, SVNIT Surat, Sep 2026*.
Hackathons: IBM Hackathon 2025 National Finalist (24-hour national round); AutonomousHacks '26, GDG Gandhinagar — offline finalist, Agentic AI track (**2,000+ participants**); CodeFest — 2nd place, Arka Jain University.
Leadership: General Secretary & President, Code & Compute Society; Community Lead, GDG on Campus.